YOU LAB · DATA-AUGMENTATION PROJECT

Expanding the augmenter's training data
Results update

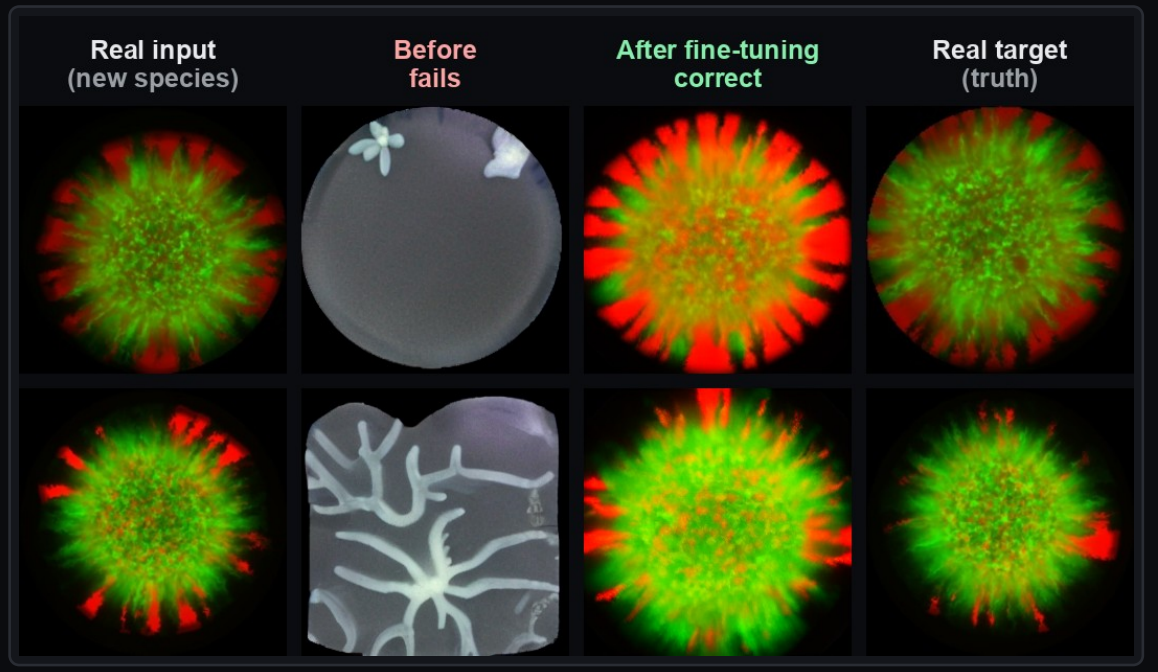
Finding more replicate data, fine-tuning on a new P. aeruginosa dataset, and what broadening the data does — and doesn't — do.

Sarthak · progress update · September 2026

RECAP — WHERE WE WERE

The model adapts to new data via fine-tuning

- Established last update: the augmenter is realistic in-distribution but fails out-of-distribution (imposes its trained pattern).
- Fine-tuning on the new domain FIXES it — on the 2-species set, realism went 2.5× → 1.16× the real-vs-real baseline.
- So the plan became: get MORE real replicate data + adapt the model to it.



Prior result (2-species): before → after fine-tuning → real target

THE DATA SEARCH

Real replicate data of our morphology is scarce

- Ran three verified search passes across Zenodo, figshare, Dryad, BioImage Archive, IDR, Kaggle, Mendeley, HuggingFace and paper supplements.
- Genuine open datasets with our dendritic/branching morphology and real replicates barely exist.
- The one real SAME-SPECIES replicate set found: GaTech Rattray-2023 — P. aeruginosa colonies, ~77 strains × 4 biological replicates, background-removed, ~256px. Staged + documented.
- Most other open 'microbial spatial growth' data is off-morphology (swarms, biofilms, fungi) or behind sign-up walls.

Takeaway: the biggest untapped lever is our own internal P. aeruginosa branching data — no open substitute exists.

RESULT — NEW DATASET

Fine-tuning on Rattray adapts to the target

2.18× → 1.69×

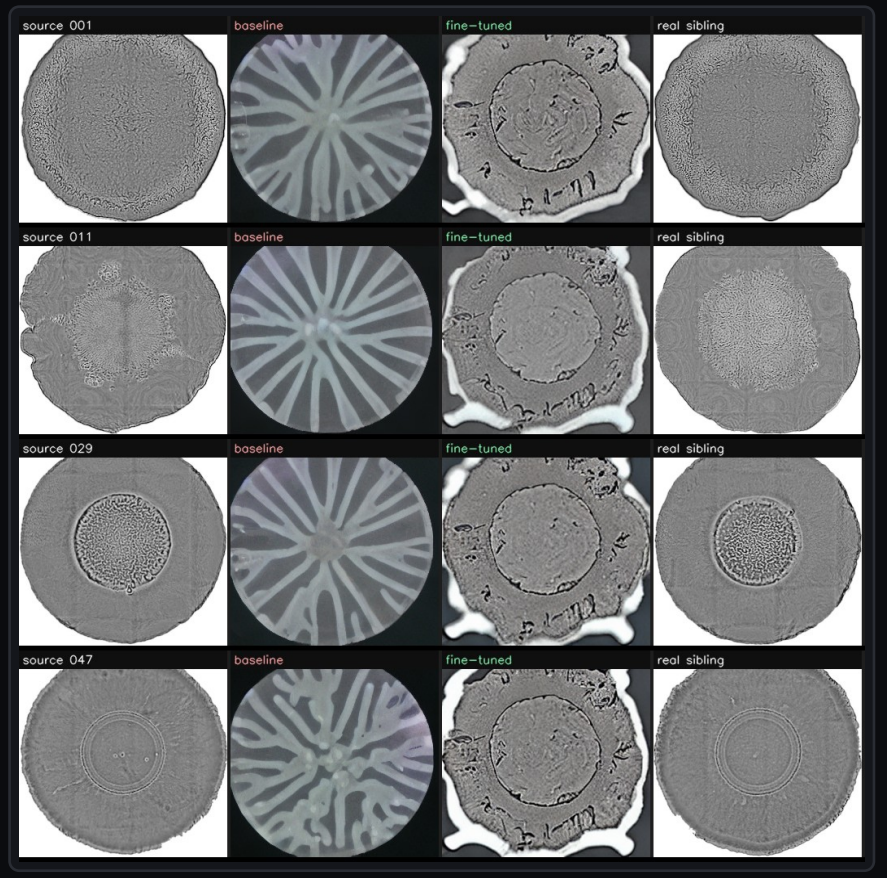
Realism vs real-replicate baseline (1.0× = ideal)

306 → 226

FID on held-out Rattray (lower = better)

Real same-species adaptation.

- Held-out test: 13 strains reserved, zero leakage.
- Baseline imposes a radial pattern; the fine-tuned model produces compact colonies matching the real replicates.



source | baseline | fine-tuned | real sibling

METHODOLOGY & RIGOR

How we tuned it

1	**Held-out test**	13 strains reserved (zero leakage); metric = realism_vs_sibling ÷ real-vs-real baseline.
2	**Layer freezing REVERSED**	Full fine-tune beat freezing here — freezing helps for a LARGE domain shift (2-species) but hurts NEAR-domain (Rattray).
3	**Epoch curve**	Optimal ≈ epoch 3; more epochs don't help (mild overfit after).
4	**Bayesian optimization**	Built an Optuna/TPE search (lr × freeze × epochs × control) — confirmed the optimum region efficiently; reusable for bigger sweeps.

RESULT — DOES MORE DATA HELP?

Broadening on all of SwarmEvo did not help

Broadened the base on ALL 47,625 SwarmEvo pairs (every frame, 81 colonies), then fine-tuned on Rattray. Scored on the held-out Rattray test (lower = better):

Rattray-only (direct fine-tune)	1.69× realism — best
Broaden → Rattray (all 47.6k SwarmEvo)	1.82× realism — worse
Broadened base alone	1.92× realism
Baseline (no fine-tune)	2.18× realism

Why: *SwarmEvo is a different organism/morphology (Enterobacter swarm) — broadening shifts the base away from the P. aeruginosa target faster than fine-tuning recovers it.*

THE DATA LESSON

More data helps — only if it's on-target

The augmenter is faithful only in-distribution (now confirmed four independent ways):

Fine-tune on target species (2-species, Rattray)	Works → adapts to the new data
Inference-parameter tuning	No effect on the OOD failure
Downstream sim→exp augmentation	Naive injection hurts
Broadening on off-morphology data (SwarmEvo)	Doesn't help — slightly hurts

So the productive expansion is TARGET-DOMAIN data (more P. aeruginosa colonies) — not volume of distant data.

NEXT STEPS & DISCUSSION

Get more target-domain data

Recommended

- Internal EMBO / Luo P. aeruginosa branching replicates — exact morphology, real replicates, lab-owned. The 'more data' that should actually help.
- Full-resolution Rattray from the authors (public copy is 256px-downsampled).
- Hold off on more distant open datasets — evidence says they won't help.

Questions for you

- Can we get the raw full-res EMBO/Luo branching replicate images (via Nan Luo)?
- Worth emailing the Rattray authors for full-resolution originals?
- Is 'efficient adaptation to a new dataset' the right near-term goal to present?